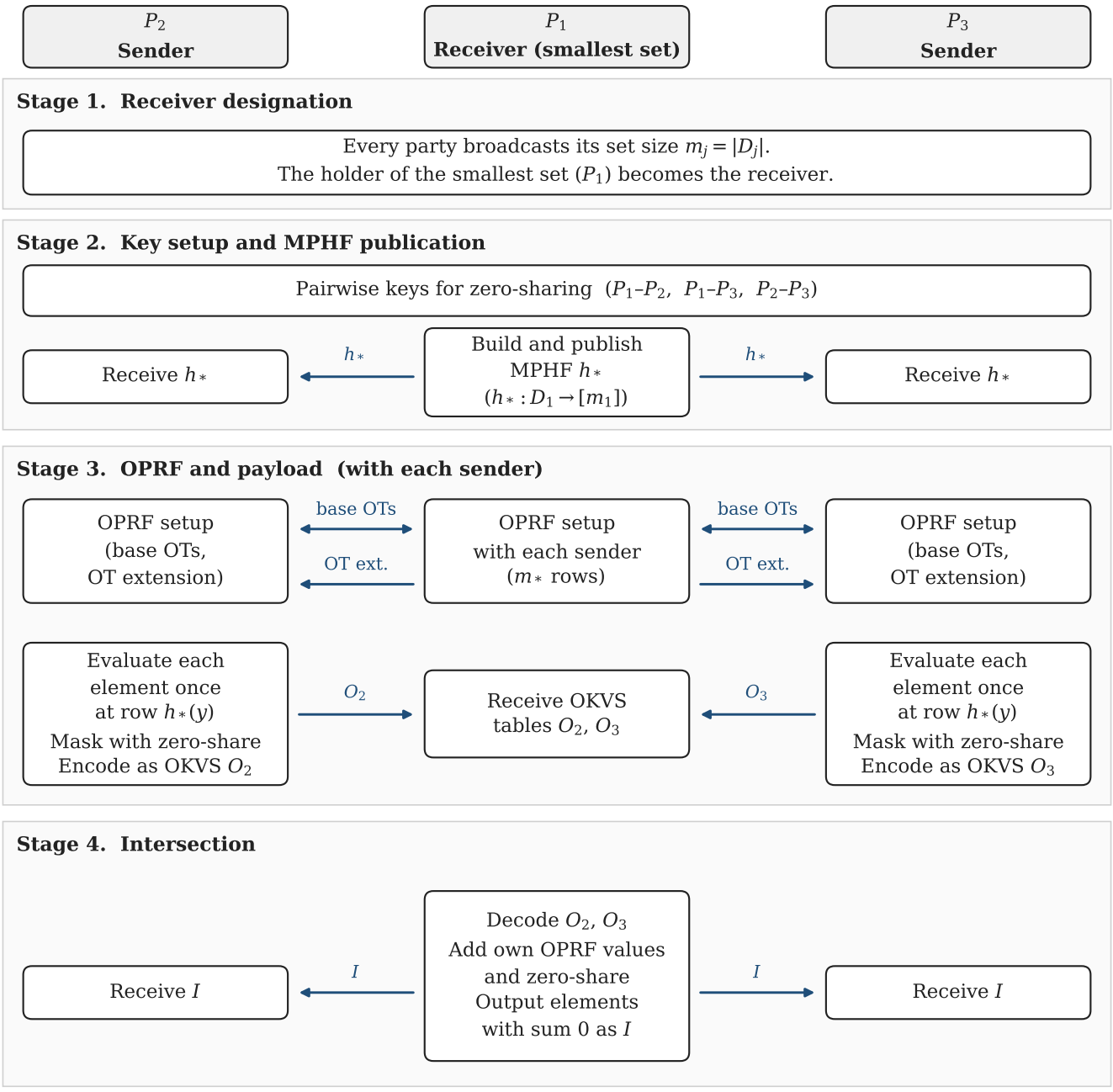
P_2
Sender
P_1
Receiver (smallest set)
P_3
Sender
Stage 1. Receiver designation
Every party broadcasts its set size $m_j = |D_j|$.
The holder of the smallest set (P_1) becomes the receiver.
Stage 2. Key setup and MPHF publication
Pairwise keys for zero-sharing (P_1–P_2, P_1–P_3, P_2–P_3)
Receive h_*
h_*
Build and publish MPHF h_*
($h_* : D_1 \to [m_1]$)
h_*
Receive h_*
Stage 3. OPRF and payload (with each sender)
OPRF setup (base OTs, OT extension)
base OTs
OT ext.
OPRF setup with each sender (m_* rows)
base OTs
OT ext.
OPRF setup (base OTs, OT extension)
Evaluate each element once at row $h_*(y)$
Mask with zero-share
Encode as OKVS O_2
O_2
Receive OKVS tables O_2, O_3
O_3
Evaluate each element once at row $h_*(y)$
Mask with zero-share
Encode as OKVS O_3
Stage 4. Intersection
Receive I
I
Decode O_2, O_3
Add own OPRF values and zero-share
Output elements with sum 0 as I
I
Receive I